

Name: Ahmad Ali Saifan

Associate Professor in Software Engineering

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Education

Ph.D.	2010, Queen's University, Canada
Dissertation Title	Runtime Conformance Checking of Mobile Agent Systems Using Executable Models
M.Sc	2003, Yarmouk University, Jordan
Emphasis	Computer Science
B.Sc.	2001, Yarmouk University, Jordan
Major	Computer Science

Professional Experience

Associate Professor. Information Systems Department, Faculty of Information Technology and Computer Sciences, Yarmouk University, Irbid, Jordan (Sep 2019-).

Associate Professor. Computer Information Systems Department, Faculty of Information Technology and Computer Sciences, Yarmouk University, Irbid, Jordan (Sep 2018- Sep 2019).

Associate Professor. Software Engineering Department, Faculty of Information Technology and Computer Sciences, Yarmouk University, Irbid, Jordan (Sep 2016- Sep 2018).

Deputy Director of the Language Center, language center, Yarmouk University, Irbid, Jordan. (Sep 2018- Sep 2019).

Director of the Language Center, language center, Yarmouk University, Irbid, Jordan (Jan 27, 2019- Feb 3, 2019)

Assistant director for academic affairs quality assurance, language center, Yarmouk University, Irbid, Jordan. (Aug 2017- Sep 2018)

Chairman of the Software Engineering Department, Faculty of Information Technology and Computer Science, Yarmouk University, Irbid, Jordan. (Sep-2016- Aug 2017).

Assistant Professor. Computer Information Systems Department, Faculty of Information Technology and Computer Sciences, Yarmouk University, Irbid, Jordan (Sep-2010 – Sep 2016)

Chairman of the Computer Information Systems, Faculty of Information Technology and Computer Science, Yarmouk University, Irbid, Jordan. Sep-2011- Sep. 2013.

Research and Teaching Assistant, School of Computing, Queen's University, Ontario, Canada (2007- 2008).

Instructor – Computer Information Systems Department, Faculty of Information Technology and Computer Sciences, Yarmouk University, Irbid, Jordan (Sep-2003 - May 2006)

Research Interests

- Software Engineering
- Software testing and verification
- Mining Software repositories
- model-based testing
- Model checking
- Mobile agent systems

Publications

❖ Journal Articles

- **Ahmad A. Saifan**, Zainab Lataifeh (2020). Privacy Preserving Defect Prediction Using Generalization and Entropy-based Data Reduction. **Intelligent Data Analysis (IDA)**. (Accepted).
- **Ahmad A. Saifan**, Lana Obiedat (2020). Feature Location Enhancement Based On Source Code Augmentation With Synonyms Of Terms. Software: Practice and Experience. Accepted. (ISI).

- **Ahmad A. Saifan** , Lina Abuwardih (2020). Software Defect Prediction Based on Feature Subset Selection and Ensemble Classification. ECTI Transactions on Computer and Information Technology (ECTI-CIT). accepted (scopus).
- **Ahmad A. Saifan**, Ahmad Alzyoud (2020) Mutation Testing to evaluate Android Applications. International Journal of Open Source Software and Processes 11(1), article 2. (scopus).
- **Ahmad A. Saifan**, Nawzat Alsmadi (2019). Source Code-Based Defect Prediction Using Deep Learning and Transfer Learning. Journal of Intelligent Data Analysis, 23(6). (ISI , IF= 0.691)
- Omar Meqdadi, Nouh Alhindawi, Jamal Alsakran, **Ahmad Saifan**, Hatim Migdadi (2019). Mining software repositories for adaptive change commits using machine learning techniques. Information and Software Technology Journal, 109(2019). (ISI, IF= 2.627).
- **Ahmad A. Saifan**, Mahmoud Bani Ata (2019). Mutation Testing for Evaluating PHP Web Applications. International Journal of Software Innovation. Vol 7(4) article 3. (scopus)
- Izzat Asmadi, Ahmed Aleroud, **Ahmad A. Saifan** (2018). Fault-based Testing for Discovering SQL Injection Vulnerabilities in Web Applications. International Journal of Information and Computer Security (IJICS). (scopus)
- Emad Alsukhni, Ahmed Aleroud, **Ahmad A. Saifan** (2018). A hybrid Pre-Post Constraint-based Framework for Discovering Multi-dimensional Association Rules Using Ontologies, International Journal of Information Technology and Web Engineering, Volume 14, Issue 1, Article 6.
- Mohammed Akour, Iman Al Jarrah, **Ahmad A Saifan** (2018). An Efficient Approach For Test Suite Reduction Using K-Means Clustering. Journal of Theoretical & Applied Information Technology 96(17). (scopus)
- **Ahmad A. Saifan**, H Alsghaier, K Alkhateeb (2017). Evaluating the Understandability of Android Applications. International Journal of Software Innovation (IJSI) 6 (1), 44-57. (scopus)

- **Ahmad A. Saifan**, Mahmoud Bani Ata, Bilal Abul-Huda (2019). Investigating The Applicability of Generating Test Cases for Web Applications Based on Traditional Graph Coverage. International Journal of Computer Aided Engineering and Technology, Vol.11 No.3. (scopus)
- Emad Alsukhni, **Ahmad A. Saifan**, Hanadi Alawneh (2017). A new data mining-based framework to test case prioritization using software defect prediction. International Journal of Open Source Software and Processes 8(1), article 2. (scopus)
- Feras Hanandeh, **Ahmad A. Saifan** , Mohammed Akour, Noor Al-Hussein, Khadijah Shatnawi (2017). Evaluating Maintainability of Open Source Software: A Case Study. International journal of open source software and processes. Volume 8 issue 1. (scopus)
- Akour, M., **Ahmad A. Saifan**, and Ratha'an O, (2016). Traceability between Code and Design Documentation in Database Management Systems: A Case Study, The International Arab Journal of Information Technology (IAJIT), (ACCEPTED. To be published in volume 15 No 5. (ISI)
- **Ahmad A. Saifan**, Emad Alsukhni, Hanadi Alawneh, Ayat Al-Sbaih, (2016). Test Case Reduction using Data Mining Technique, International Journal of Software Innovation, (ACCEPTED). volume 4 issue 4: 56-70. (scopus)
- **Ahmad A. Saifan**, Akour M., Alazzam, I., Hanandeh F. (2016). Regression Test-Selection Technique Using Component Model Based Modification: Code to Test Traceability. International Journal of Advanced Computer Science and Applications (IJACSA). 7(4): 90-95. (scopus)
- **Ahmad A. Saifan** and Wafa Bani Mustafa, Using Formal Methods For Test Case Generation According To Transition-Based Coverage Criteria, Jordanian Journal of Computers and Information Technology. Dec 2015, 1(1), pp. 15-30.
- Mohammed G. Al-Zamil, S. Samarah, **Ahmad A. Saifan**, I. Al Smadi: Dispersion-based prediction framework for estimating missing values in wireless sensor networks. International Journal of Sensor Networks (IJSNet), 12(3): 149-159 (2012).
- Samer Samarah, Mohammed Al Zamil, **Ahmad A. Saifan**.

Model Checking Based Classification Technique for Wireless Sensor Networks. New Review of Information Networking 17(2):93-107. 2012.

- Eslam Al Maghayreh, Samer Samarah, Faisal Alkhateeb, Iyad Abu Doush, Izzat Alsmadi, and **Ahmad A. Saifan**, A Framework for Monitoring the Execution of Distributed Multi-agent Programs, International Journal of Advanced Science and Technology, Vol. 38, January, 2012, pp. 53-66.
- Izzat Alsmadi, Samer Samarah, **Ahmad A. Saifan** and Mohammed G. AL Zamil. Automatic Model Based Methods to Improve Test Effectiveness, Universal Journal of Computer Science and Engineering Technology, 1(1), 41 - 49. 2010.
- Izzat Alsmadi , Samer Samarah , **Ahmed A. Saifan**, and Mohammed AL Zamil. Evaluate and Improve GUI Testing Coverage Automatically. Int.J. of Software Engineering, IJSE Vol.4(2):41-60 July 2011
- Ghassan Kanaan, Riyad Al-Shalabi, Mohammed G. AL-Zamil, **Ahmad A. Saifan**: Comparison between Ad-hoc Retrieval and Filtering Retrieval Using Arabic Documents. Int. J. Comput. Proc. Oriental Lang. 17(3): 181-199 (2004).

❖ **Conferences**

- **Ahmad A. Saifan**, Areej Alrabadi, Evaluating Maintainability of Android Applications (2017). The 8th International Conference on Information Technology ICIT 2017. Amman, Jordan.
- Hind Husni, **Ahmad A. Saifan**, Cloud Testing: Steps, Tools, Challenges (2017), New Trends in Information Technology – NTIT 2017, Amman, Jordan.
- **Ahmad A. Saifan**, Test Case Reduction using Data Mining Classifier Techniques. (2016). 2nd International Conference on Computer and Information Technology (ICCIT 2016). Istanbul, Turkey April 11-12, 2016.
- **Ahmad A. Saifan**,. and Wahsheh, H. A. 2012. Mutation

operators for JADE mobile agent systems. In Proceedings of the 3rd international Conference on information and Communication Systems (Irbid, Jordan, April 03 - 05, 2012). ICICS '12. ACM, New York, NY, 1-5. DOI=<http://doi.acm.org/10.1145/2222444.2222460>.

- Nesreen Al-Khalidi, **Ahmad A. Saifan** and Izzat Alsmadi. Selecting a Standard Set of Attributes for Cost Estimation of Software Projects, 2012 International Conference on Computer, Information and Telecommunication Systems, CITS 2012, Amman, Jordan.
- **Ahmad A. Saifan**, Jürgen Dingel, Jeremy S. Bradbury, Ernesto Posse: Implementing and Evaluating a Runtime Conformance Checker for Mobile Agent Systems. In Software Testing, Verification and Validation (ICST), 2011 IEEE Fourth International Conference on (pp. 269-278). IEEE.
- **Ahmad A. Saifan**, Ernesto Posse, Jürgen Dingel: Run-time conformance checking of mobile and distributed systems using executable models. In Proceedings of the 7th Workshop on Parallel and Distributed Systems: Testing, Analysis, and Debugging (PADTAD) (p. 1). ACM. 2009.
- **Ahmad A. Saifan**, Jürgen Dingel: A Survey of Using Model-Based Testing to Improve Quality Attributes in Distributed Systems. In Advanced Techniques in Computing Sciences and Software Engineering (pp. 283-288). Springer Netherlands.

❖ Books

- Riyadh Al-Shalabi, Ghassan Kanaan, Mohammad Rawwash and **Ahmad A. Saifan**. Applications in Computer Sciences and statistics. 2005. 300pp.

❖ Other Publications

- **Ahmad A. Saifan**, " Runtime Conformance Checking of Mobile Agent Systems Using Executable Models ", PhD thesis, Queen's University, 2010.

- **Ahmad A. Saifan** and Juergen Dingel. Model-based testing of distributed systems. Technical Report 2008-548, School of Computing, Queen's University, 2008.

Funded Projects

- Suileman H. Mustafa, Qasem A. Al-Radaideh, **Ahmad A. Saifan**, Mohammed A. Akour, Iyad M. Al-Azzam. Joint programs and framework for doctoral education in software engineering - THE «PATHWAYS TO PHD» - PWS@PHD. Erasmus+. (60285€), 2015.
- **PI: Ahmad Saifan.** Approved initial proposal of a funded project from the **Scientific Research Support Fund** has been accepted. The detailed proposal has been submitted on April 26, 2016 with a total amount of **60000 JD** (\$83333).

Honors & Awards

- ❖ Awarded as one of the best 10 software specification projects in Zayed university competition in IT, April 2011.
- ❖ Awarded the eleventh place as a coach in the Jordanian collegiate Programming Contest for the unexpected team from Yarmouk university, August 2015 .
- ❖ Awarded a scholarship from Yarmouk University (in 2001) to complete my Master degree in Computer Science.
- ❖ Awarded a scholarship from Yarmouk University (in 2004) to complete my PhD degree in Computer Science.
- ❖ Queen's Graduate Award. Queen's University, Ontario, Canada. 2007
- ❖ Queen's General Bursary. Queen's University, Ontario, Canada. 2006-2009

Editorship and Refereeing Activities

- Reviewer in the journal of software: evolution and process 2019-now
- Reviewer in the International Journal of Open Source Software and Processes (IJOSSP) 2017-now.
- Reviewer in the International Arab Journal of Information Technology 2010-now
- Reviewer in International Journal of Advanced Computer Science and Applications. IJACSA
- Technical committee member in International Conference on Computer and Information Technology (ICCIT 2016)

- Program committee member in the 7th International Conference on Computer Science and Information Technology (CSIT 2016)
- Program committee member in the New Trends in Information Technology conference – NTIT 2017

Other Academic Activities

- Chair of the ACM programming competition group, Faculty of Faculty of Information Technology and Computer Science, Yarmouk University. For the year 2016/2017.
- Chair and Member of the Curriculum Committee, Department of Computer Information systems, Yarmouk University, for the academic years 2011/2012 and 2012/2013. And member for academic year 2015/2016.
- Member of the quality assurance committee at the Faculty of Information Technology and Computer Science, Yarmouk University, for the academic year 2012/2013.
- Member of the Investigation Committee for Students Violations at the Faculty of Information Technology and Computer Science, Yarmouk University, for the academic year 2014/2015.
- Chair and Member of the Software Engineering/Bachelor Program Committee, Department of Computer Information Systems, Yarmouk University, for the academic year 2011/2012.
- Chair and Member of the Software Engineering/Master Program Committee, Department of Computer Information Systems, Yarmouk University, for the academic year 2014/2015.
- Chair and Member of the Graduate Studies Committee, Department of Computer Information Systems , Yarmouk University, for the academic year 2011/2012, 2012/2013.
- Member of the Quality assurance Committee, Faculty of Information Technology and Computer Sciences, Yarmouk University, for the academic year 2012/2013.

Teaching Experience

- Yarmouk University 2003-2006, 2010- till now. I have taught the following courses:
 - CIS 100 Computer Skills 2.
 - CIS 103 Introduction to Information Technology.
 - CIS 104 Introduction to Information Systems.
 - CIS 110 Internet Programming.
 - CIS 240 Software Engineering.
 - CIS 256 File Structure.

- CIS 342 System Analysis and Design
- CIS 440 Object Oriented analysis and design.
- CIS 499 Bachelor Graduation Project
- CIS 642 Software Testing and Performance Assessment.
- CIS 641 Advanced Software Engineering.
- CIS 646 Advanced Object-Oriented analysis and Design.
- CIS 695 Master Graduation Project

Supervisor for the following master thesis:

- Test case reduction using deep learning, by Raghad Drabseh. From December 2019-now.
- Test Case Reduction Techniques: An Empirical Comparison Study, by Haya Alquraan, From October 2019-now.
- Privacy Preserving Defect Prediction Using Generalization and Entropy-based Data Reduction, by Zainab Lataifeh, from August 2018-now.
- Feature location enhancement based on source code augmentation with synonyms of terms, by Lana Obeidat, from August 2018- August 2019
- Source Code-based Defect Prediction using Deep Learning and Transfer Learning, by Nawzat Alsmadi, from July 2017- April 2018.
- Identifying Logical Errors Locations in Database Management Systems using Event Extraction, by Osama Ratha'an. December 2016.
- Mutation Testing to Evaluate Android Applications, by Ahmad Al-Zyoud, July 2016
- A classifier to detect profit and non profit websites upon textual metrics, by Dirar Darweesh, co-supervised with Dr. Yahya Tashtoush. JUST. December 2014.
- Investigating the Ability of Testing Web Application Based on Traditional Software Testing Techniques, by Mahmmoud Bani Ata. December 2015 .

Graduation Project Supervision at Yarmouk University

- Lina Abu Wardih
- Wafa Bani Mustafa
- Heba Badarneh
- Nesreen Al khalidi

- Rasha al Jarrah

Examiner for the following thesis:

- Cloud based Breast Cancer Diagnostic using CNN, by Samih Qasem, 2020 Al al-Bayt University
- Using Ensemble Paradigm in Testcase Reduction by Mohammad Bani Domi, 2019, Yarmouk university
- Analysis and Improvement of Grouping-Proof Protocols for Radio Frequency Identification Systems, by Halah Sami Abu Dhaileh, 2018 JUST.
- An empirical inspection of the refactoring tools impacts on software quality, by Shatha shlool, 2015, JUST.
- Traceability enhancements on source code analysis tools to improve software defects prediction, by Qusai Al-Zoubi, 2013, Yarmouk University.
- A notional understanding of the relations between code readability and software complexity, by Noor Abu El Rub, 2013, JUST.
- Sentimental Information Retrieval for Arabic Social and News Networks, by Safaa Bani Essa, 2013, JUST.

Membership of Committees

- IEEE member
- Member of the IEEE computer Society
- Member of the Graduate Studies Committee, Department of Computer Information Systems , Yarmouk University, for the academic year 2011/2012, 2012/2013.
- Member of the Quality assurance Committee, Faculty of Information Technology and Computer Sciences, Yarmouk University, for the academic year 2012/2013.

References

Prof. Dr. Samer Samarah, CIS Department, Faculty of IT, Yarmouk university, e-mail: samers@yu.edu.jo

Prof. Dr. Mohammad Alzamil, CIS Department, Faculty of IT, Yarmouk university, e-mail: mohammedz@yu.edu.jo

Prof . Dr. Bilal Anas Hamed Abul-huda, Dean of faculty of Information Technology and Computer Sciences, Yarmouk university, e-mail: abul-huda@yu.edu.jo

Prof. Dr. Sulieman Mustafa, CIS Department, Faculty of IT, Yarmouk university, e-mail:
mustafa@yu.edu.jo